

Fractions and Binomial

John Doe

June 24, 2026

The binomial coefficient, $\binom{n}{k}$, is defined by the expression:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

1. Displaying fractions

Fractions can be used inline within the paragraph text, for example $\frac{1}{2}$, or displayed on their own line, such as this:

$$\frac{1}{2}$$

2. Text-style fractions

We use the `amsmath` package command `\text{...}` to create text-only fractions like this:

$$\frac{\text{numerator}}{\text{denominator}}$$

Without the `\text{...}` command the result looks like this:

$$\frac{numerator}{denominator}$$

3. Size and spacing within typeset mathematics

Fractions typeset within a paragraph typically look like this: $\frac{3x}{2}$. You can force L^AT_EX to use the larger display style, such as $\frac{3x}{2}$, which also has an effect on line spacing. The size of maths in a paragraph can also be reduced: $\frac{3x}{2}$ or $\frac{3x}{2}$. For the `\scriptscriptstyle` example note the reduction in spacing: characters are moved closer to the *vinculum* (the line separating numerator and denominator).

Equally, you can change the style of mathematics normally typeset in display style:

$$f(x) = \frac{P(x)}{Q(x)} \quad \text{and} \quad f(x) = \frac{P(x)}{Q(x)} \quad \text{and} \quad f(x) = \frac{P(x)}{Q(x)}$$

Continued fractions

Fractions can be nested but, in this example, note how the default math styles, as used in the denominator, don't produce ideal results...

$$\frac{1 + \frac{a}{b}}{1 + \frac{1}{1 + \frac{1}{a}}}$$

...so we use `\displaystyle` to improve typesetting:

$$\frac{1 + \frac{a}{b}}{1 + \frac{1}{1 + \frac{1}{a}}}$$

Here is an example which uses the `amsmath \cfrac` command:

$$a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \cdots}}}$$

Here is another example, derived from the `amsmath` documentation, which demonstrates left and right placement of the numerator using `\cfrac[l]` and `\cfrac[r]` respectively:

$$\frac{1}{\sqrt{2} + \frac{1}{\sqrt{2} + \frac{1}{\sqrt{2} + \cdots}}}$$

Example

$$a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \cdots}}}$$